

CoughSense : Arduino Tiny ML-Based Real-Time Cough Detection

Problem Statement

Cough is a key early symptom of respiratory infections (COVID-19, influenza, TB), yet continuous monitoring outside clinical settings is nearly impossible. Cloud-based and mobile apps introduce unacceptable latency, transmit sensitive audio data, and drain battery, making them unsuitable for passive home monitoring.

Our Goal: Deploy a lightweight, privacy-preserving cough detector that runs entirely on an Arduino Nano 33 BLE Sense - no data ever leaves the device.

Data

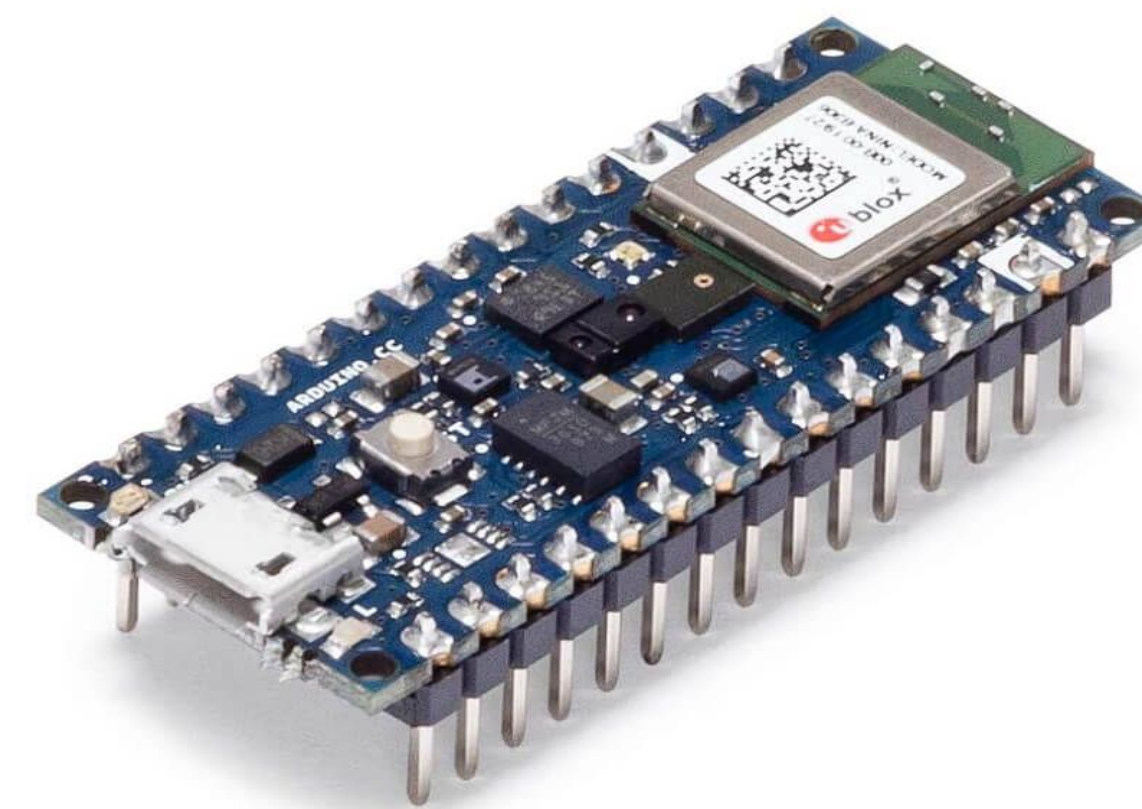
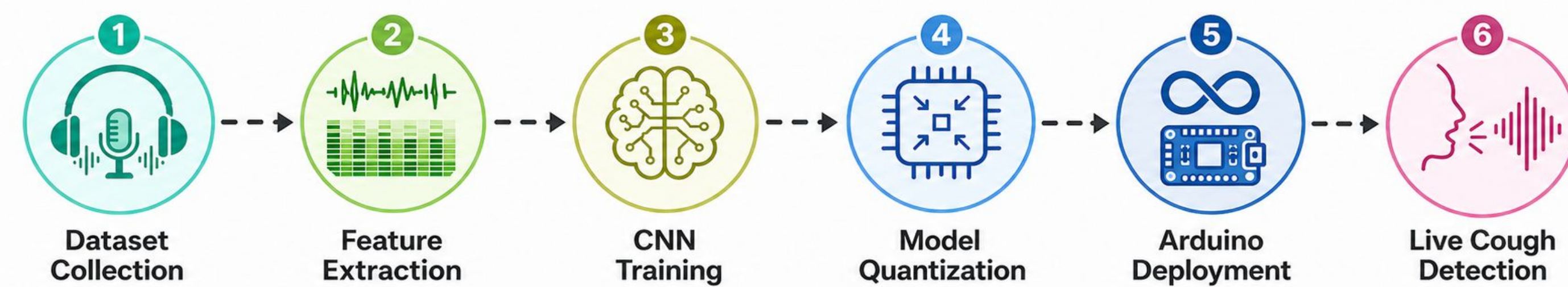
Coswara Dataset - Respiratory audio repository: cough, speech, breathing, noise. [IISc Bangalore]

Processing - Cough → positive; Others → non-cough; Standardized audio, removed noisy samples.

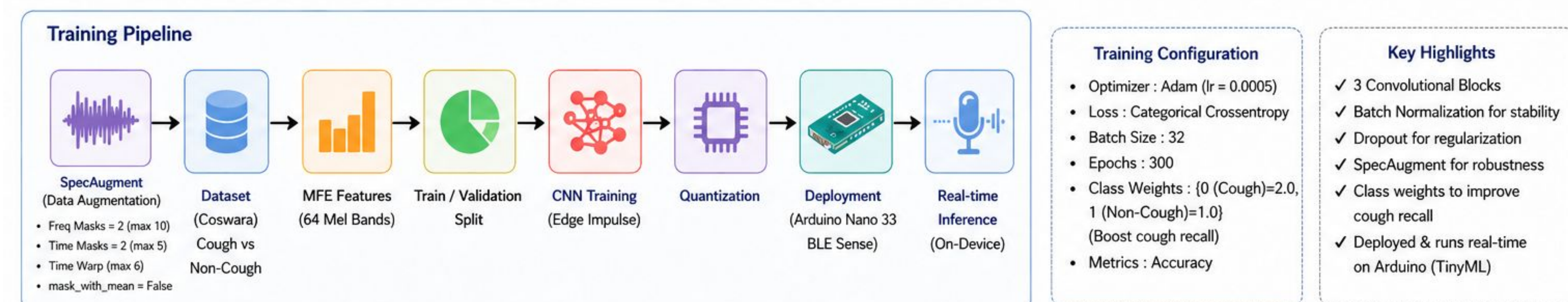
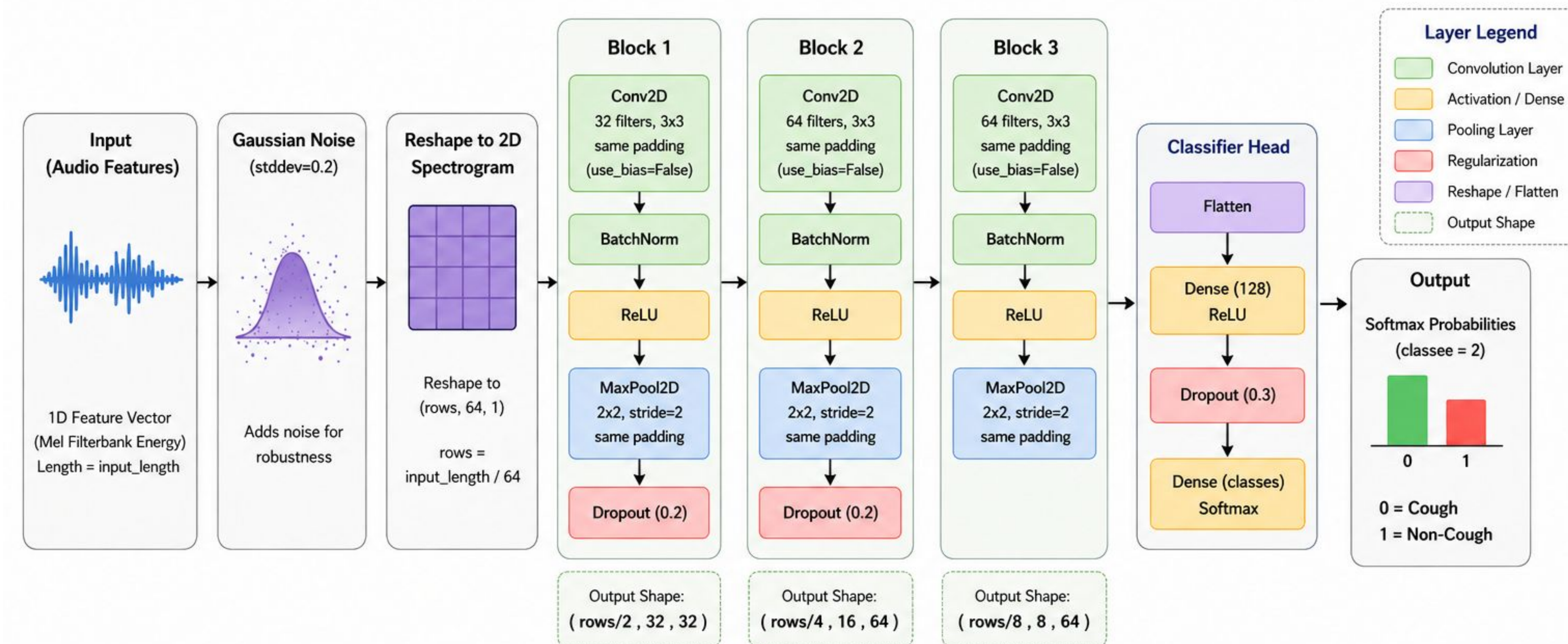
Window Size: 1,000 ms with a **200 ms** stride **Frequency: 16,000 Hz**

MFE Configuration: 64 filters, 25ms/10ms frame/stride, and 512 FFT points

Approach



Network Architecture



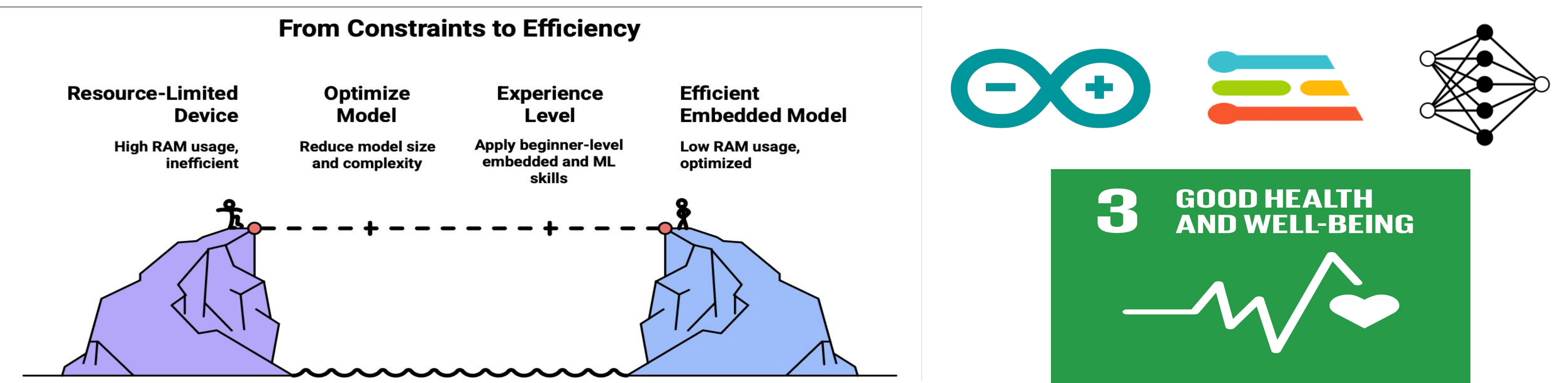
Note: input_length = length of MFE feature vector (multiple of 64). rows = input_length / 64

CoughSense Neural Network Architecture

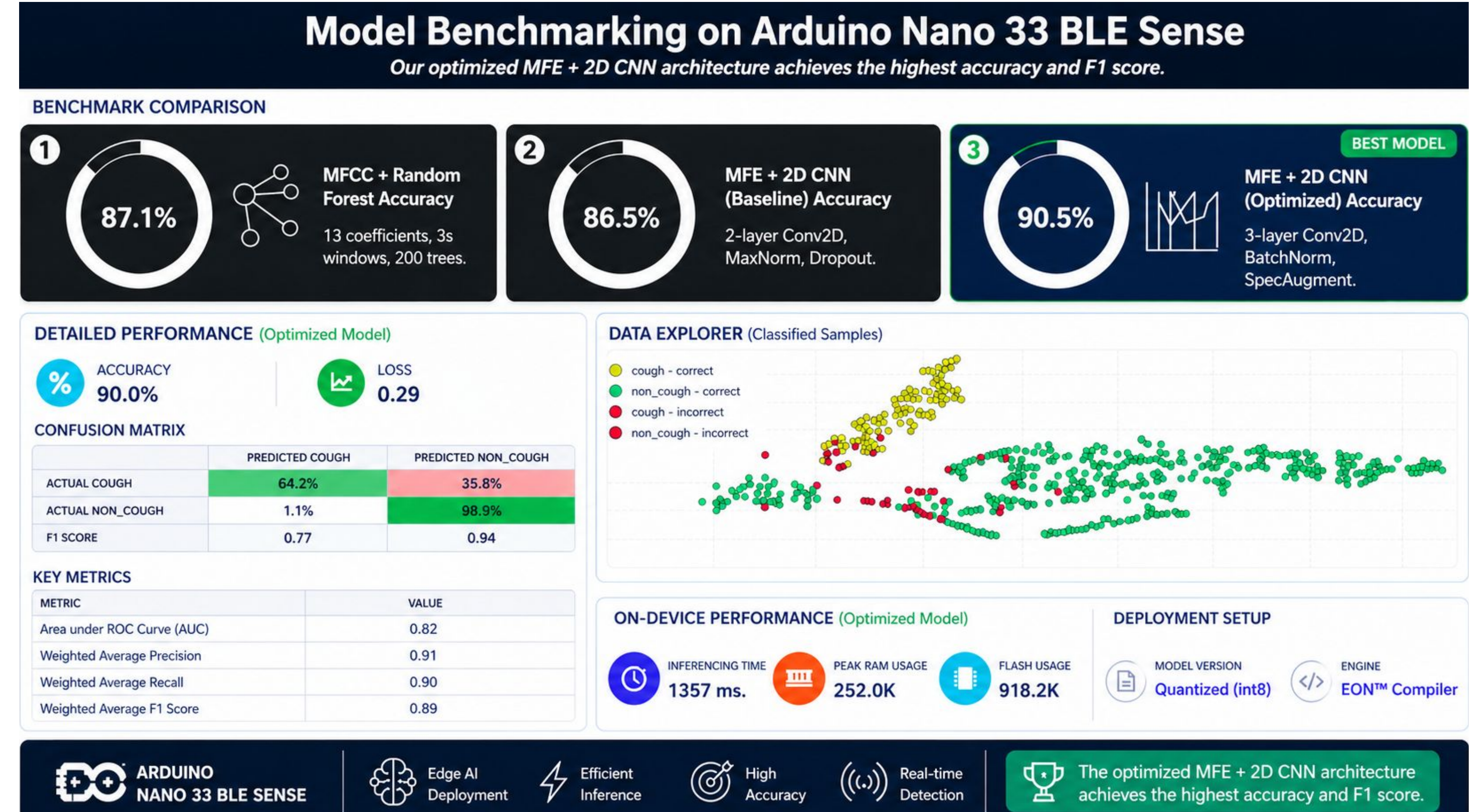
Implementation & Constraints

CoughSense is a privacy-first, real-time cough detection system deployed entirely on an Arduino Nano 33 BLE Sense microcontroller using TinyML and EdgeImpulse - no cloud, no latency, no data transmission.

- On-device inference in **1.5s** using only **125.7 KB** RAM and **234.3 KB** Flash.
- Mel Filterbank Energy (MFE) features optimized for microcontroller DSP.
- 3-layer 2D CNN achieving **90.5%** accuracy with **F1 = 0.89**.
- SpecAugment + noise augmentation for robust real-world performance.



Evaluation Plan and Results



Main References

- [1] Bhattacharya, D. et al. "Coswara: A respiratory sounds and symptoms dataset for remote screening of SARS-CoV-2 infection." Zenodo, Oct. 2022. DOI: 10.5281/zenodo.7188627
- [2] Edge Impulse Documentation - Arduino Nano 33 BLE Sense. (<https://docs.edgeimpulse.com/docs/edge-ai-hardware/mcu/arduino-nano-33-ble-sense>)